Online Shoppers Purchasing Intention Prediction

Project Proposal — Assignment 1

Problem Statement and Objective

Online shopping platforms collect information about visitor browsing behavior, including pages viewed, time spent on different page types, visitor characteristics, and session-related information. However, not every browsing session results in a purchase. Predicting whether a visitor will complete a purchase can help online businesses better understand customer behavior and support data-driven decision-making. The objective of this project is to develop a machine learning classification model that predicts whether an online shopping session will result in a purchase. The target variable, Revenue, indicates whether a session resulted in a purchase (True) or did not result in a purchase (False).

Dataset and Target Variable

The project uses the Online Shoppers Purchasing Intention dataset, which contains 12,330 online shopping sessions and 18 variables. After separating the target variable, 17 predictor features are used. These include numerical measures such as page visits, browsing duration, bounce and exit rates, page values, operating systems, browsers, regions, and traffic types, as well as categorical variables such as Month, VisitorType, and Weekend. The dataset is suitable for this problem because its variables describe visitor behavior and session characteristics that can be used to investigate purchasing outcomes. Revenue is a binary target, making the problem a binary classification task.

Data Preparation and Pipeline

The dataset was examined for missing values, duplicate records, potential outliers, and class imbalance. No missing values were found. Duplicate records were identified and considered during data-quality analysis. Potential outliers were investigated using the IQR method but were not automatically removed because extreme browsing behavior may represent legitimate sessions. The target classes were found to be imbalanced, so appropriate evaluation metrics will be used rather than relying on accuracy alone. An 80/20 stratified train-test split is used. Numerical features are processed with median imputation and StandardScaler, while categorical features are processed with most-frequent imputation and OneHotEncoder. These transformations are implemented through scikit-learn ColumnTransformer and Pipeline so that preprocessing parameters are learned only from the training data, preventing test-set information from leaking into training.

Evaluation and Success Metric

The initial model is Logistic Regression. Model performance will be assessed using precision, recall, F1-score, and a confusion matrix, with particular attention to F1-score and recall for the purchasing class because the target is imbalanced. This will provide a more informative assessment of the model's ability to identify purchasing sessions than accuracy alone.